

SynthID Forensic Analysis Report

A (Yours): your_output.jpg | B (Theirs): working_tool_output.jpg

1. File & Metadata

Property	Yours (A)	Theirs (B)
File size	1499.77 KB	685.26 KB
Dimensions	1440x1440	1440x1440
Format	JPEG	JPEG
EXIF tags present	0	0

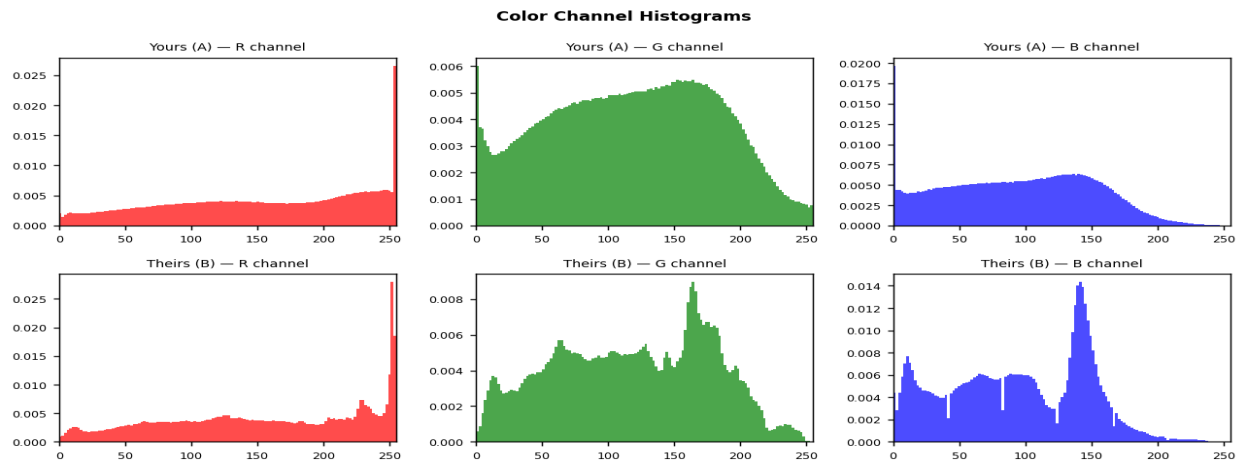
■ File size differs by 814.5 KB. Larger file often means less aggressive JPEG compression was applied — which preserves more of the original pixel structure including the watermark.

2. Pixel-Level Difference Statistics

Metric	Value	Meaning
Mean absolute diff	21.4234 / 255	Average per-pixel change. Target: 3–8
Max diff	244.0 / 255	Largest single pixel change
Std of diff	27.7805	Spread of changes across image
PSNR	19.25 dB	Quality loss. >35dB = invisible, <25dB = visible
R channel mean diff	-2.6524	
G channel mean diff	-0.1084	
B channel mean diff	0.8086	

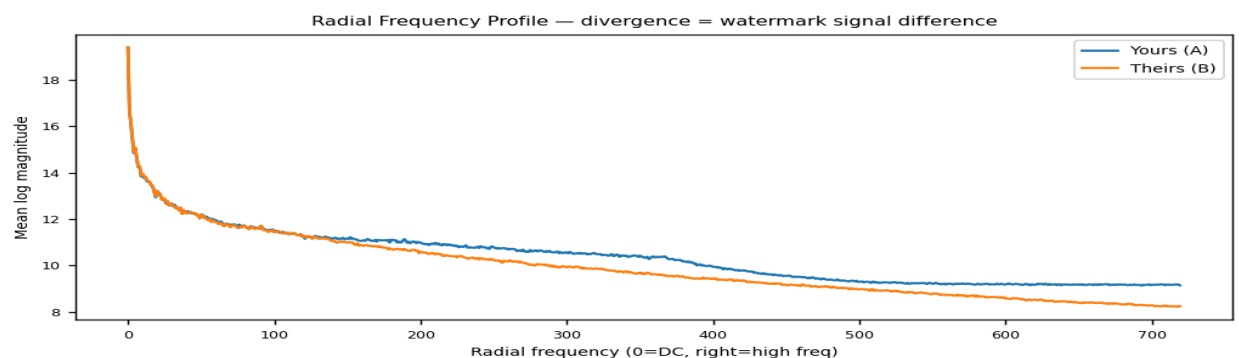
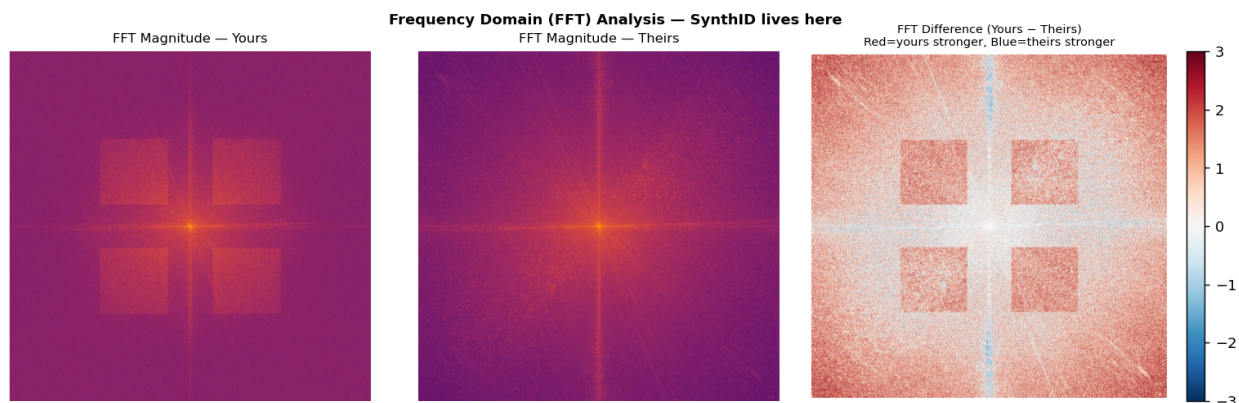
3. Color Channel Histograms

Differences in histogram shape between A and B indicate colour processing differences. SynthID removal tools often subtly shift channel distributions as a side effect of their perturbations.



4. Frequency Domain (FFT) Analysis

This is the most important section. SynthID is embedded in the mid-frequency bands of the image. The FFT magnitude maps show where each image has energy. The difference map shows which tool disrupted more frequency content — and where. The radial profile shows the exact frequency bands each tool is targeting.



FFT Metric	Yours (A)	Theirs (B)
Mean magnitude	9.6163	8.9099
Std magnitude	0.9705	1.1401
Mid-freq diff (A-B)	0.9846	↑ higher = more disruption
Overall FFT diff	0.9720	↑ higher = more disruption

5. DCT Coefficient Analysis

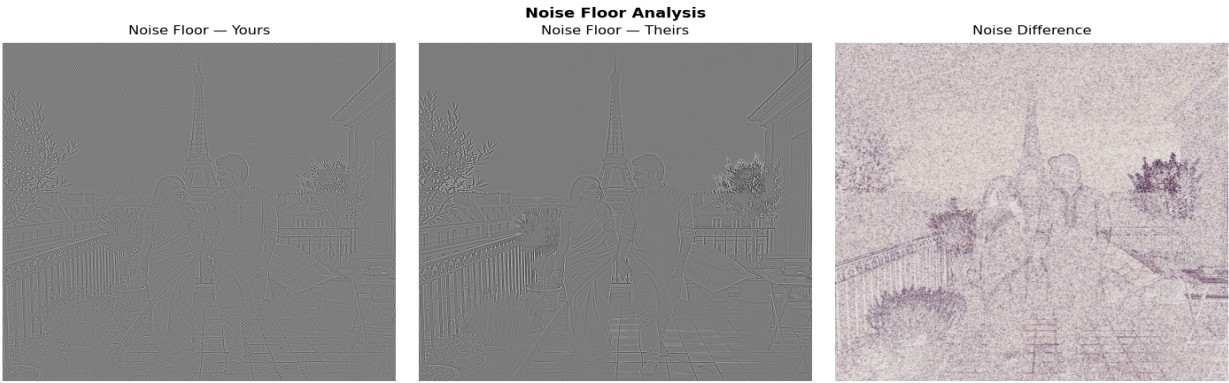
JPEG and SynthID both operate in the DCT domain. SynthID modifies mid-range DCT coefficients. If the working tool is disrupting different DCT bands than yours, that gap is likely why their removal works and yours doesn't.



DCT Band	Diff magnitude	Notes
Low freq (DC area)	47.784016	Should be LOW — don't touch DC
Mid freq (target)	34.366558	Should be HIGH — this is SynthID's home
High freq	7.720836	Less important for SynthID
Overall mean	14.275818	

6. Noise Floor Analysis

The noise floor shows what each tool leaves behind after processing. SynthID tools typically add a calibrated noise layer — too little and the watermark survives, too much and the image looks grainy.



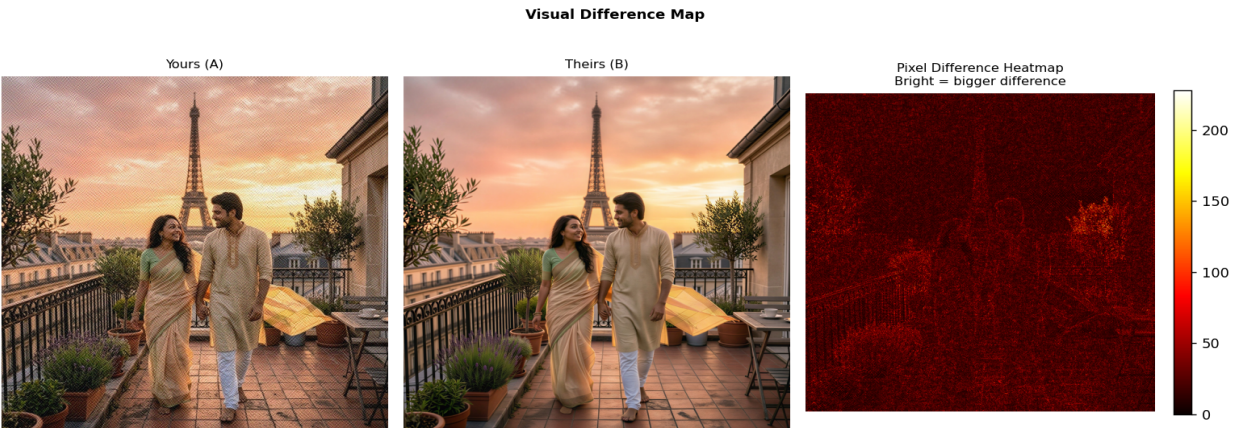
Noise Metric	Yours (A)	Theirs (B)
Noise std dev	14.6910	7.5242
Mean abs noise	11.6388	3.5821
Noise diff std	15.3338	↑ = different noise patterns

7. Texture & Entropy Analysis

Metric	Yours (A)	Theirs (B)	Notes
Entropy	7.9046	7.8173	Higher = more randomness
Contrast	0.6294	0.2150	Local intensity variation
Homogeneity	0.7507	0.9208	Smoothness of texture
Energy	0.2431	0.3523	Uniformity of texture
Correlation	0.8989	0.9633	Linear dependency of grey levels

8. Visual Difference Heatmap

Bright areas in the heatmap show where the two images differ most. A good SynthID removal tool produces changes spread evenly across the entire image — not concentrated in corners or specific regions.



9. Key Findings & Recommendations

- **PIXEL DIFF:** Large pixel difference (21.42/255). Your processing is too aggressive — causing visible quality loss.
- **FFT MID-FREQ:** Good mid-frequency disruption (0.9846). This is the target range for SynthID removal.
- **DCT COEFFS:** Low-freq DCT change (47.7840) > mid-freq (34.3666). You are changing the wrong coefficients — SynthID is in mid-range DCT, not low-freq.

■ **NOISE FLOOR:** Your noise (14.691) is much higher than theirs (7.524). You are adding too much noise causing graininess.

■ **ENTROPY:** Entropy similar (yours 7.905, theirs 7.817).

What to change in your script:

1. Reduce filter strength. You want pixel diff in the 3–8 range — enough to disrupt signal, not enough to see.
2. Your DCT perturbation is hitting the wrong bands. Target DCT indices $[h//8:h//2, w//8:w//2]$ — avoid the DC and very-low coefficients.
3. Reduce your noise injection strength — you are over-noising the image.

This report is for technical research into digital watermarking systems.